Daily Residual Statistical Arbitrage in Recent Technology Equities Replication

Thiago Amin

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1 Objective and Scope

Pairs trading begins with the idea that economically similar securities should not diverge persistently after accounting for their common exposure. The generalized version replaces a single peer stock with one or more factor portfolios. A stock is bought when it appears abnormally cheap relative to its factor-implied value and sold when it appears abnormally expensive, provided the residual behavior is sufficiently consistent with mean reversion.

This replication asks whether that daily residual mean-reversion mechanism is detectable in modern U.S. technology equities. The initial study will:

- operate only at the daily trading frequency;
- use adjusted daily prices and traded ETF factors;
- estimate signals from trailing windows without future information;
- evaluate realized PNL after transaction costs; and
- report results for a recent fixed sample, proposed as January 2019 through December 2025.

The first implementation is not intended to establish that daily trading is the optimal horizon. It is a single-horizon baseline that can later become one component of a multi-horizon allocation study.

2 Data and Study Universe

2.1 Technology Stock Universe

Let $\mathcal{U}_t$ denote the investable stock universe on trading day t. The target universe is liquid U.S.-listed technology-sector equities. To avoid survivorship bias in a formal replication, membership should be constructed from point-in-time holdings of a technology benchmark ETF or another point-in-time sector classification. A simplified exploratory implementation may begin with a fixed set of liquid technology stocks, but results from that variant must be labeled as conditional on the selected universe rather than as a survivorship-free sector study.

For each stock $i \in \mathcal{U}_t$, let $S_{i,t}$ be its split- and dividend-adjusted closing price. Daily log returns are

$$R_{i,t} = \log\left(\frac{S_{i,t}}{S_{i,t-1}}\right).$$

Adjusted data are required because a mechanical dividend or split should not be interpreted as an idiosyncratic pricing dislocation.

2.2 Factor Benchmarks

The primary factor specification follows the paper's interpretable ETF approach. Because the trading universe is restricted to technology equities, the primary factor is the traded technology-sector benchmark:

$$F_t = R_{\mathrm{XLK},t}.$$

The stock is therefore assessed relative to the sector portfolio against which it is economically comparable.

A pre-specified robustness analysis may use multiple traded benchmark returns, for example

$$F_t = \begin{bmatrix} R_{\mathrm{SPY},t} & R_{\mathrm{XLK},t} & R_{\mathrm{QQQ},t} \end{bmatrix}^\top.$$

This specification tests whether separating broad-market, technology-sector, and technology-heavy growth exposure materially changes the residual signal. Because these ETFs may be highly correlated, the single-factor XLK model remains the primary clean replication benchmark.

3 Residual Relative-Value Model

3.1 Factor Decomposition

For one factor, the daily stock return is decomposed as

$$R_{i,t} = \alpha_i + \beta_i F_t + \widetilde{R}_{i,t}.$$

The systematic component is $\beta_i F_t$ and $\widetilde{R}_{i,t}$ is the idiosyncratic residual return. Under a multi-factor specification,

$$R_{i,t} = \alpha_i + \sum_{j=1}^{m} \beta_{ij} F_{j,t} + \widetilde{R}_{i,t}.$$

The research focus is not solely whether the factors explain return variance; it is whether the remaining residual contains a stable and tradable mean-reversion signal.

If a dollar exposure $Q_{i,t}$ is taken in stock i, the associated factor-hedged position takes ETF exposures

$$Q_{F_j,t} = -\beta_{ij} Q_{i,t}, \qquad j = 1, \ldots, m.$$

For that stock-level trade, estimated factor exposure is neutral:

$$\beta_{ij} Q_{i,t} + Q_{F_j,t} = 0.$$

Aggregating these hedged stock trades produces a daily long–short portfolio designed to earn residual convergence rather than broad technology-market direction.

3.2 Residual Level

Let M be the number of observations in the trailing estimation window. In line with the source study, the baseline is

$$M = 60 \text{ business days}.$$

At a decision date t, all parameters are estimated using observations ending no later than $t-1$. Let $\widehat{\widetilde{R}}_{i,u}$ be the residual return from the fitted factor regression for $u = t-M, \ldots, t-1$. Define the cumulative residual level over the estimation window by

$$X_{i,u} = \sum_{v=t-M}^{u} \widehat{\widetilde{R}}_{i,v}.$$

The residual level represents the cumulative relative-value displacement of the stock from its factor hedge. A large positive value indicates accumulated outperformance relative to the benchmark; a large negative value indicates accumulated underperformance.

4 Mean-Reversion Estimation

4.1 Ornstein–Uhlenbeck Specification

For each stock, the residual level is modeled as an Ornstein–Uhlenbeck process:

$$dX_i(u) = \kappa_i \left(m_i - X_i(u)\right) du + \sigma_i dW_i(u), \qquad \kappa_i > 0.$$

Here m_i is the long-run residual equilibrium, κ_i is the speed of mean reversion, and σ_i is residual volatility. The conditional expected residual change is

$$\mathbb{E}_t[dX_i(u)] = \kappa_i \left(m_i - X_i(u)\right) du.$$

Thus a residual above equilibrium predicts negative relative return, while a residual below equilibrium predicts positive relative return.

4.2 Discrete Daily Estimation

With $\Delta = 1/252$ year, the exact daily discretization can be estimated as an AR(1) regression:

$$X_{i,u+\Delta} = a_i + b_i X_{i,u} + \eta_{i,u+\Delta}.$$

The OU interpretation requires

$$0 < b_i < 1.$$

For an accepted fit, the estimates map to mean-reversion quantities through

$$\widehat{\kappa}_i = -\frac{\log(\widehat{b}_i)}{\Delta}, \qquad \widehat{m}_i = \frac{\widehat{a}_i}{1-\widehat{b}_i}.$$

If $\widehat{\sigma}^2_{\eta,i}$ is the variance of the AR(1) innovation, the estimated equilibrium standard deviation of the residual level is

$$\widehat{\sigma}_{\text{eq},i} = \frac{\widehat{\sigma}_{\eta,i}}{\sqrt{1-\widehat{b}_i^2}}.$$

The estimated characteristic mean-reversion time is

$$\widehat{\tau}_i = \frac{1}{\widehat{\kappa}_i}.$$

The source strategy restricts trading to residuals that revert sufficiently quickly relative to the 60-day estimation window. The baseline eligibility rule for this replication will therefore be

$$\widehat{\tau}_i < \frac{30}{252}, \qquad \text{equivalently} \qquad \widehat{\kappa}_i > \frac{252}{30}.$$

Stocks failing this rule, failing the stationary AR(1) restriction, or lacking sufficient data are not eligible for a new position on that date.

5 Trading Signal

5.1 S-Score

The standardized residual displacement at decision date t is

$$s_{i,t} = \frac{X_{i,t-1} - \widehat{m}_i}{\widehat{\sigma}_{\mathrm{eq},i}}.$$

This score measures how far the stock's factor-relative residual lies from its estimated equilibrium in equilibrium-standard-deviation units. The use of $X_{i,t-1}$ reflects the implementation convention that a position for day t may use only information observable before that day's return is realized.

5.2 Stateful Entry and Exit Rules

The baseline daily signal retains the threshold structure reported in the source study:

$$\begin{array}{lll}
\text{buy stock / sell hedge to open} & \text{if} & s_{i,t} < -1.25, \\
\text{sell stock / buy hedge to open} & \text{if} & s_{i,t} > +1.25, \\
\text{close short-stock position} & \text{if} & s_{i,t} < +0.75, \\
\text{close long-stock position} & \text{if} & s_{i,t} > -0.50.
\end{array}$$

The strategy is stateful: an entry signal opens a position and the corresponding exit rule determines when it is unwound. It is intentionally a discrete baseline rather than a continuously scaled function of the s-score, so that the replication first tests the source trading mechanism.

6 Portfolio Construction and PNL

Let $z_{i,t} \in \{-1, 0, +1\}$ denote the active stock-side signal for stock i on day t, where $+1$ is long and -1 is short. Let Λ_t be the dollar amount assigned to each active stock-side position per unit of current portfolio equity E_t. Stock and hedge dollars are

$$Q_{i,t} = E_t \Lambda_t z_{i,t}, \qquad Q_{F_j,t} = -E_t \Lambda_t \sum_{i \in \mathcal{U}_t} z_{i,t} \widehat{\beta}_{ij,t}.$$

The implementation will choose Λ_t so that portfolio gross exposure does not exceed a pre-specified budget. Both stock-side and ETF hedge exposures must be included in this gross-exposure accounting.

Let w_t denote the complete vector of stock and ETF portfolio weights applied over day t, and let R_t denote their realized daily return vector. With transaction cost rate c, daily net PNL in return units is

$$\pi_t = w_t^\top R_t - c \left\| w_t - w_{t-1} \right\|_1 .$$

The timing convention is conservative: w_t is fixed using information available through day $t-1$ and earns only returns observed on day t. To connect to the paper, the benchmark transaction-cost setting is

$$c = 0.0005,$$

or five basis points per dollar of changed position. Results should also be reported under lower and higher cost assumptions because modern liquidity and implementation quality differ from the historical sample.

7 Estimation and Backtest Protocol

For each out-of-sample trading date t:

1. Construct the eligible technology-stock universe using only point-in-time information available at t.
2. Obtain adjusted daily returns for stocks and factor ETFs through date $t-1$.
3. Using the trailing 60 business days, fit the factor regression for each stock and form its residual-level series.
4. Fit the AR(1)/OU model and retain only stocks satisfying the stationarity and mean-reversion-time eligibility rules.
5. Compute each eligible stock's s-score and update its open or closed position state.
6. Construct the corresponding ETF hedges and rescale positions to the gross-exposure limit.
7. Apply those positions to the return realized on date t, deduct turnover costs, and record net PNL.

All model fitting and signal construction are rolling and out of sample. The choice of universe, factors, 60-day window, thresholds, exposure budget, and transaction-cost assumptions must be fixed before inspecting performance in the test period. If alternative thresholds or factor sets are explored, they will be reported as separate specifications rather than silently chosen from the best backtest result.

8 Results To Report

The eventual implementation report should include:

- universe construction and daily counts of eligible/traded stocks;
- factor specification and distributions of estimated factor loadings;
- distributions of $\widehat{\kappa}_i$, $\widehat{\tau}_i$, and s-scores;
- number of entries, exits, average holding period, and turnover;
- cumulative net PNL, annualized volatility, annualized Sharpe ratio, maximum drawdown, and hit rate;
- average long gross, short gross, hedge gross, net exposure, and factor exposure after hedging;
- transaction-cost sensitivity; and
- comparison of the primary XLK factor model against pre-specified factor robustness variants.

9 Secondary Replication Extension: PCA Factors

The original study also extracts systematic returns through principal components. This is a useful second-stage replication, but it is not required for the initial ETF implementation. For a trailing window, standardized stock returns can be collected in a matrix Y and the empirical correlation matrix formed as

$$\rho = \frac{1}{M-1} Y Y^{\top}.$$

If $v^{(1)}, \ldots, v^{(k)}$ are eigenvectors corresponding to the retained largest eigenvalues, their associated eigenportfolio returns become estimated factors. The residual/OU/s-score/backtest procedure then remains unchanged; only the source of the systematic factor returns changes. Comparing ETF and PCA specifications within the same recent technology universe will test whether the residual strategy is sensitive to how systematic technology risk is extracted.

10 Implementation Sequence

1. Implement adjusted daily price and ETF return ingestion.
2. Implement point-in-time technology universe construction, with any simplified fixed-universe fallback clearly labeled.
3. Implement rolling ETF regression residualization.
4. Implement OU parameter estimation, eligibility screening, and s-score computation.
5. Implement stateful factor-hedged positions, exposure limits, and transaction-cost-adjusted PNL.
6. Produce the tables and plots specified above.
7. Only after the daily replication is established, expose this strategy as a one-day sleeve for the endogenous-horizon study.

Source Study

Marco Avellaneda and Jeong-Hyun Lee, *Statistical Arbitrage in the U.S. Equities Market*, July 11, 2008, SSRN abstract 1153505.